



Welcome & “What Is a Computer Thinking?”

Modern Chinese School · Summer 2026 AI & Coding · Session 1 Worksheet

Mon Jun 29, 2026 · Ages 7–

13

Phase 0 · Week 1

☆ TIER A ♦ TIER B 🦋 TIER C

✓ TEACHER ANSWER SHEET — SESSION 1 WORKSHEET

NAME

Teacher Copy — Do Not Distribute

SECTION

1 / 2 / 3

DATE

Jun 29, 2026

MY TIER

A / B / C

📖 PART 1 · LESSON SUMMARY — KEY CONCEPTS

💡 **Big Idea:** A computer can only do exactly what it's told — one step at a time, in order. AI is the same: it follows patterns and rules. Better instructions = smarter result.

👤 **AI Moment:** Today's demo — you give a “robot” step-by-step instructions to make a sandwich. Computers do exactly what they're told. AI still just follows instructions and patterns.

📖 **Teacher Note:** These are the reference definitions. Accept any accurate paraphrase. Award full credit if the core idea is correct.

Algorithm

A precise, ordered list of steps that solves a problem — like a recipe for a computer

✓ **Key: ordered + precise + solves a problem**

Instruction

One single step — a computer does each one exactly, in order, no guessing

✓ **Key: ONE step, exact, no guessing**

AI

Programs that learn from patterns in data to make guesses or suggestions

✓ **Key: learns from patterns in data**

Bug

A mistake in the steps — bugs are normal! Programmers find and fix them every day

✓ **Key: mistake in steps — normal!**

🔧 PART 2 · IN-CLASS ACTIVITY — MODEL ANSWERS

🔥 Warm-Up — Robot Shoes

Accept any logical sequence. Model answers per tier below.

☆ TIER A — 3 Steps (draw OR write)

- 1 Pick up right shoe with both hands.
- 2 Slide right foot all the way inside the shoe.
- 3 Pull the velcro strap tight and press down.

♦ TIER B — 5 Precise Steps (which hand? which foot?)

- 1 Use right hand to pick up left shoe from the floor.
- 2 Use left hand to open the shoe opening wide.
- 3 Point left foot's toes down and push foot inside.
- 4 Heel must touch the back of the shoe — push down.
- 5 Grab both laces and tie a bow knot tightly.

🦋 TIER C — 5 Steps + if/else (wrong foot?)

- 1 Look at right shoe — find the bump on the inside sole.

🔧 Guided Build — Sandwich Algorithm

Model answer below. Groups may vary — reward precision and logical order.

- 1 Take 2 slices of bread from the bag and place on the plate.
- 2 Open the jar of peanut butter — use the knife to scoop 1 tablespoon.
- 3 Spread peanut butter on one side of one bread slice, covering all edges.
- 4 Press the second slice on top, unspread-side down. Done!

⚠ **Most Common Trouble Step:** Step #3 — “spread to cover edges” is vague to a robot. Reward groups that specified “all the way to the crust.”

🔧 Open Build — My Own Algorithm

Sample task: Brushing Teeth. Accept any daily routine with correct tier structure.

☆ TIER A — 4 Steps (draw + label)

- 2 Hold shoe with both hands, opening facing you.
- 3 **if** right shoe → push right foot in heel-first.
else swap shoes and repeat Step 1.
- 4 Check: can you wiggle toes? **if** NO → loosen.
if YES → tie laces.
- 5 Cross laces, loop, pull tight, form bow.

Vocab Match — Answer Key

1. **Algorithm** → **D** Precise ordered list of steps
2. **Instruction** → **B** One single step
3. **AI** → **C** Programs that spot patterns
4. **Bug** → **A** A mistake in steps

- 1 Get toothbrush and toothpaste from the cup.
- 2 Squeeze a pea-size dot of toothpaste onto brush.
- 3 Brush all teeth in small circles for 2 minutes.
- 4 Spit toothpaste out, rinse mouth with water.

◆ **TIER B — 6 Steps + 1 if**

- 1- Same as Tier A, more precise (exact amounts and actions).
- 4 **if** toothpaste runs out → get a new tube from cabinet.
- 6 Rinse toothbrush, put back in cup upright.

TIER C — 6+ Steps + 2 if/else

- 1- All Tier B steps, with added precision (“30 seconds per quadrant”).
- 6 **if/else** **if** brush is dry → wet it first. **else** proceed directly.
- if/else** **if** still tastes dirty → brush again 1 min. **else** rinse and finish.
- 2

EXIT TICKET — MODEL ANSWERS

An algorithm is...

✓ **A precise, ordered list of steps that tells a computer exactly what to do — like a recipe. Every step matters because computers follow instructions literally.**

One AI in my life...

✓ **Accept: Siri/Alexa (voice AI), Netflix/YouTube recommendations (pattern AI), spam filter, autocorrect, Face ID. Explain what pattern it recognizes.**

A question I still have...

✓ **Open-ended — all thoughtful questions are valid! Great examples: “Can AI make its own algorithm?” / “What happens if a step is missing?” / “Does AI ever feel confused?”**

PART 3 • HOMEWORK (~30 MIN) • DUE TUE JUN 30 — ANSWER KEY

 Plan: 0–5 min: read • 5–20 min: scavenger hunt • 20–28 min: algorithm • 28–30 min: exit question

AI Scavenger Hunt — Sample Answers

Any real-world AI is valid. Award full credit for accurate description of what the AI does.

1 — I found AI on/in:

✓ **e.g. “My phone’s autocorrect” — This AI predicts the next word I type based on patterns in how people write.**

2 — I found AI on/in:

✓ **e.g. “Netflix” — The AI watches what shows I pick and suggests new ones it thinks I’ll like.**

3 — Surprise find:

Write Your Algorithm — Answer Key

Sample Task: Feeding a Pet (Cat)

☆ **TIER A — 4 Steps**

- 1 Go to kitchen and find cat food bag.
- 2 Scoop one cup of dry food into bowl.
- 3 Place bowl on floor in cat’s eating spot.
- 4 Call the cat by name to come eat.

◆ **TIER B — 6 Steps + 1 if (bonus)**

- 5 **if** water bowl is empty → fill with fresh water from tap.
- 6 Wait 3 seconds — call cat again if it didn’t come.

✓ e.g. “The thermostat” — It learns what temperature we like and adjusts itself automatically.

🔧 TIER C — 6-8 Steps + 2 if/else + real AI decision

if/else `if` bag is empty → go to pantry for backup. `else` scoop directly.

1

if/else `if` cat doesn't eat after 5 min → check if sick. `else` clean bowl.

2

Real AI Decision: “A smart feeder AI checks the cat's weight and automatically dispenses less food if gaining weight — pattern-based decision making!”

💬 One Last Thought

After the scavenger hunt, I was surprised that AI...

✓ “...is in my refrigerator! It tracks how long food has been inside. I thought AI was only in phones.”

- ✓ Found 3 AI examples at home
- ✓ Wrote what each AI does (1 sentence each)
- ✓ Completed my tier's algorithm
- ✓ Name + tier written at top ✓